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Introduction

Introduction to Linear Models

Reading the Data

Implementing Linear Models for Image Classification

Neural Networks and Deep Neural Networks for Image Classification

Deep Neural Networks with Dropout and Batch Normalization

Video: Deep Neural Networks with Dropout and Batch Normalization
7 min

Video: Lab Introduction - Classifying Images using Dropout and Batchnorm Layer
11 sec

Graded App Item: Lab: Classifying Images using Dropout and Batchnorm Layer
1h 30m

Reading: Readings
30 min

Quiz: New Quiz
8 min

Introduction

Convolutional Neural Networks

Understanding Convolutions

CNN Model Parameters

Working with Pooling Layers

Implementing CNNs on Vertex AI by using a pre-built TF container

Introduction

Preprocessing the image data

Model parameters and the data scarcity problem

Data Augmentation

Transfer Learning

Summary

New Quiz

✔ Submit your assignment

Due Feb 11, 11:59 PM IST

✔ Receive grade

To Pass 80% or higher

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✔ Congratulations! You passed!

Grade received	Latest Submission Grade	To pass	Go to next item
100%	100%	80% or higher	

			Try again
1.	What does the loss function do?	1 / 1 point	
The loss function computes the updated model based on the data being observed. The loss function groups layers into an object with training and inference features. The loss function computes the average value of the cost function over all the training samples. The loss function measures how accurate the model is during training.			Your grade 100% View Feedback We keep your highest score
2.	When can a sequential model be used?	1 / 1 point	
A sequential model is appropriate for a plain stack of layers where each layer has exactly one input tensor and multiple output tensors. A sequential model is appropriate for a plain stack of layers where each layer has multiple input tensors and one output tensor. A sequential model is appropriate for a plain stack of layers where each layer has exactly one input tensor and one output tensor. A sequential model is appropriate for a plain stack of layers where each layer has multiple input tensors and multiple output tensors.			✔ Correct Correct. A sequential model is not appropriate when the model has multiple inputs or multiple outputs.
3.	What function can be used for a model to do prediction?	1 / 1 point	
model.compile() model.fit() model.predict() model.evaluate()			✔ Correct Correct. The model.predict() function is used for a model to do prediction.
4.	When is the dropout technique used?	1 / 1 point	
Dropout is a technique used to prevent a model from underfitting. Dropout is a technique used to prevent a model from overfitting. Dropout is a feature added between the layers of the neural network, and it continuously takes the output from the previous layer and normalizes it before sending it to the next layer. Dropout is a technique used to remove a small percentage of weights at each iteration. So weights will never be equal to zero.			✔ Correct Correct. Dropout is a regularization technique that prevents neural networks from overfitting. During training, dropout randomly discards a portion of the neurons to avoid overfitting.
5.	How does the batch normalization work?	1 / 1 point	
Batch normalization applies a transformation that maintains the mean output close to 0 and the output standard deviation close to 0. Batch normalization applies a transformation that maintains the mean output close to 0 and the output standard deviation close to 1. Batch normalization applies a transformation that maintains the mean output close to 1 and the output standard deviation close to 1. Batch normalization applies a transformation that maintains the mean output close to 1 and the output standard deviation close to 0.			✔ Correct Correct. Batch normalization normalizes the outputs using a mean equal to 0 and a standard deviation equal to 1 ($\mu=0, \sigma=1$).